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DATA DESCRIPTOR

A dataset on occupant satisfaction with the indoor environmental quality in Belgian classrooms

Quinten Carton¹✉, Jakub Kolarik² & Hilde Breesch¹

This paper describes a dataset containing field measurement data from classrooms in Belgium. The dataset comprises of survey responses, indicating occupants' satisfaction with the indoor environmental quality (IEQ) (i.e., thermal, indoor air quality, visual, acoustic and overall IEQ conditions), measurements of the indoor environmental conditions in the classroom during heating and intermediate season, and contextual variables like occupant location in classroom. Data were collected in three educational case studies: three classrooms in a secondary school (Case 1), seven primary school classrooms (Case 2), and a university lecture room (Case 3). The full dataset consists of 6834 satisfaction assessments collected from 321 unique occupants in classrooms, merged with corresponding measurements of the classroom IEQ. The dataset benefits the research community since a longitudinal dataset of this size on occupants' satisfaction in classrooms is unavailable at this moment. The dataset can serve as a training or validation dataset for novel comfort models, a basis for statistical analyses to establish a causal model or to examine inter- and intra-individual differences in occupant satisfaction.

Background & Summary

Providing a satisfactory indoor environmental quality (IEQ) is essential for safeguarding occupants' health¹, productivity^{2,3} and overall satisfaction⁴. In the case of educational buildings, an unsatisfactory classroom IEQ can lead to learning loss among pupils and students^{5–7}. However, guaranteeing high occupant satisfaction levels is challenging due the complex combination of variables influencing the satisfaction. These variables can be environmental, personal and contextual⁸. In addition, currently used comfort models like Fanger's predicted mean vote – predicted percentage dissatisfied (PMV-PPD) model⁹ and the adaptive thermal comfort model^{10,11}, often poorly predict the actual satisfaction of the individual occupant as these models predict the average comfort preference of a large group of occupants rather than predicting the unique preference of an individual occupant. In addition, these comfort models are constructed on training data from adults, while many of the occupants in classrooms are minors¹². Therefore, further research is needed to improve the understanding of occupant satisfaction with IEQ in classrooms and to improve the accuracy of predictive comfort models¹³. Personal comfort models are a promising modelling approach for enhancing the accuracy of predicting occupant preferences with regard to the IEQ. Personal comfort modelling is an approach which aims to predict the individual preferences of occupants, rather than relying on the average preference of a large group^{14,15}. The integration of personal comfort models in the HVAC control system has the potential to increase occupant satisfaction and reduce energy use by the HVAC system¹⁶. A personal comfort model is trained using occupant feedback (e.g., thermal satisfaction, sensation or preference) and one or more environmental, physiological or behaviour-related factors^{14,17–20}. Therefore, a longitudinal dataset obtained through repeated measures data collection is required to construct a personal comfort model. In these types of data collection campaigns, occupants must evaluate their perception of satisfaction multiple times throughout the study. The study duration can vary from several days to several months. The benefit of a longitudinal dataset is that it can be used to study the impact of variables that change over time (e.g., IEQ-related variables such as temperature and CO₂-concentration) on occupants' satisfaction. Many of the public datasets such as, ASHRAE Global Thermal Comfort Database²¹ and the other longitudinal datasets^{17,22} were limited to evaluating occupants' perceptions of the thermal environment and not of the other IEQ conditions, such as the indoor air quality (IAQ), visual and acoustical conditions. A dataset constructed

¹KU Leuven, Department of Civil Engineering, Building Physics and Sustainable Design, Ghent Campus, Belgium.

²Technical University of Denmark, Department of Civil and Mechanical Engineering, Kgs. Lyngby, Denmark. ✉e-mail: quinten.carton@kuleuven.be

by Jayathissa *et al.*²³ extended the scope by include assessments of occupants' perception of the thermal, acoustic and visual conditions of an indoor space. However, an evaluation of occupants' satisfaction with the IAQ conditions was missing. Therefore, there is a lack of publicly available longitudinal datasets for classrooms that evaluate occupants' satisfaction with all four IEQ domains.

Added value and novelty of the current dataset. The current paper describes the construction of a longitudinal dataset on occupants' satisfaction with the IEQ in three educational case studies. The dataset was designed to meet the following requirements:

1. Assessments of occupants' satisfaction with the IEQ conditions: The dataset was not limited to occupants' satisfaction with the thermal environment, but also included satisfaction assessments on the IAQ, acoustic and visual conditions.
2. Simultaneous monitoring of the IEQ-related, personal and contextual variables: Measurements of various IEQ-related (e.g., temperature, relative humidity, CO₂,...), personal (e.g., clothing level, age, gender,...) and contextual variables (e.g., moment of assessment, window opening, occupants' seating location,...) were performed in order to obtain a holistic overview of the potential influencing variables on occupants' satisfaction.
3. Obtained from a field study in classrooms: The dataset represented occupants' actual response to the IEQ conditions in classrooms.
4. Longitudinal measurement campaign: The dataset resulted from a longitudinal measurement campaign in which multiple assessments (e.g., 10–20 assessments per person) were collected from the occupants. Consequently, sufficient data was available to determine occupants' unique preferences and train personal comfort models.

The dataset consists of 6834 survey responses on occupants' satisfaction with the IEQ from 321 different occupants (i.e., children, teenagers and adults), merged with corresponding measurements of the classroom IEQ. All classrooms were located in the region of Flanders (Belgium). The research community can use the presented dataset as:

- A validation dataset to evaluate novel comfort models.
- A training dataset for personal comfort models since the dataset comprises of multiple votes per occupant.
- A basis for a statistical study which aims to establish a causal model for occupant satisfaction in classrooms or to study inter- and intra-individual differences in satisfaction.
- An evaluation dataset to assess the impact of new heating, ventilation and air-conditioning (HVAC) control systems on occupants' satisfaction with IEQ. For example, the dataset can be the basis for an agent-based model representing occupant satisfaction responses to HVAC control actions in a building energy simulation study.

The relevance of the current dataset can be summarised in the following three points:

- Longitudinal feedback for personal comfort modelling: Many existing datasets collect only a few comfort votes per person. Our dataset stems from longitudinal measurement campaigns where participants were frequently asked about their IEQ satisfaction. This repeated feedback is essential for developing and validating personal comfort models, which aim to predict unique occupant preferences rather than group averages.
- Comprehensive evaluation of occupants' satisfaction with IEQ: Unlike most datasets that focus solely on thermal comfort, our dataset includes evaluations of satisfaction with thermal, IAQ, acoustic, and visual conditions, enabling holistic modelling of occupants' satisfaction with IEQ.
- Underrepresented Belgian context: The ASHRAE global thermal comfort database²¹ contains only 85 datapoints from Belgium, all collected in residential buildings. Moreover, only 597 datapoints (0.5% of the ASHRAE dataset) originate from classrooms in a climate zone similar to Flanders (Belgium). Our dataset addresses this gap by providing an extensive dataset from classrooms in this underrepresented region, offering a valuable resource for studies on IEQ in educational settings within a mixed-humid climate.

Methods

This section describes the data collection and dataset construction approach. The data were collected during three case studies: three classrooms in a secondary school (Case 1), seven primary school classrooms (Case 2), and a test lecture room at a university campus (Case 3). The three case studies varied in type of occupants: children under 12 years old in case 2, teenagers in case 1 and young adults in case 3. As a result, the satisfaction of a wide range of occupants in educational buildings was studied. Firstly, the data collection approach comprising of environmental monitoring and surveys, assessing occupants' satisfaction, are described. Secondly, it delves into the monitoring period, building and system characteristics, participants sample, monitoring equipment and survey distribution of each of the three case studies. Afterwards, the data preparation process is specified (e.g. data cleaning, variable creation and data merging).

Data collection approach. Longitudinal measurements, i.e., repeated measures of occupant satisfaction with IEQ, were performed to collect data suitable for (1) determining the main influencing variables of occupants' satisfaction with IEQ via statistical analysis, and (2) construction of predictive satisfaction models. The

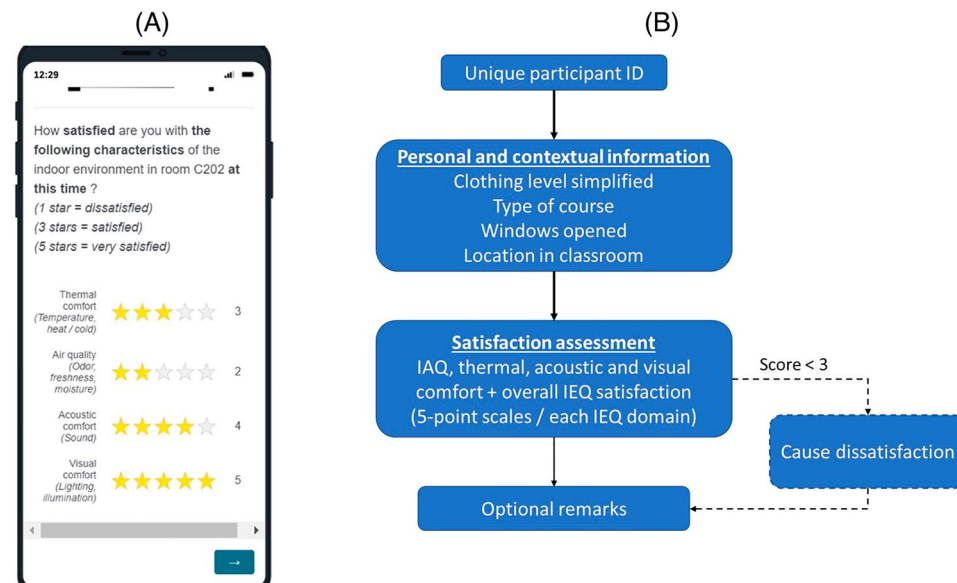


Fig. 1 Example visualisation of the RHRN survey and survey structure. **(A)** A screenshot of the satisfaction assessment via the RHRN survey on a mobile device. **(B)** The RHRN survey structure flow for case study 1.

measurement campaigns consisted of simultaneously monitoring the indoor environmental conditions in the classroom and assessing occupants' satisfaction with IEQ via online surveys.

Survey structure and content. Occupants' satisfaction was assessed via online surveys using the software Qualtrics²⁴. The surveys could be accessed by the participants via a QR-code or weblink. A two-step approach was used to survey occupants' satisfaction in the case study buildings. Firstly, a retrospective survey was distributed to gain more detailed insights on the occupants' background (e.g. demographics, typical behaviour and occupant satisfaction during past period). Secondly, shorter right-here-right-now (RHRN) surveys were completed regularly by the participants in order to assess their immediate thermal and IAQ satisfaction over a longer period. The RHRN survey started by asking participants' unique identifier which was used to track occupants' responses throughout the data collection campaigns. After the data collection, the participants' identifiers were pseudonymised to protect their privacy. Occupants' satisfaction was assessed on a five-point rating scale (see Fig. 1A). This five-point rating scale was used as it resembled a commonly used customer scale to review products and services (e.g., Google Maps²⁵, Trustpilot²⁶, Bol²⁷). Consequently, the IEQ satisfaction assessment became clear and recognisable for participants, which limited their time investment in the survey. The midpoint did not represent a neutral point and was labelled as 'satisfied' to participants. As a result, occupant responses could be categorised into 'dissatisfaction' and 'satisfaction' without neglecting neutral responses or assuming neutral responses resembled satisfaction. All four domains of the IEQ (i.e., IAQ, thermal, acoustic and visual comfort) and the overall IEQ were assessed by the participants. Scores <3 were categorised as dissatisfied, and scores ≥ 3 as satisfied. In case of a score lower than 3, a follow-up question appeared in which the participant could indicate the reason for his/her dissatisfaction.

Depending on the case study, different questions were posed before or after the satisfaction assessments. These questions asked about personal and contextual factors (e.g., clothing level, location in the classroom or window opening) or aimed to assess occupants' study engagement (Case 3) or survey fatigue (Case 2). Figure 1B shows the structure of the RHRN survey distributed in a secondary school (Case 1). A detailed overview of the content and structure of the surveys used in each case study can be found in the survey contents file at the data repository²⁸.

Informed consent procedure. At the start of each data collection campaign an informed consent procedure was performed. The procedure started with informing the participants were about the content, aim and usage of the surveys. Afterwards, their consent to participate with the data collection campaign was asked at the start of the first survey they completed. In addition, the participants consented to their data being used for scientific purposes and publication, provided that their privacy and anonymity were protected. Additional measures were taken in case the participants were minors. An informed consent form was provided to a parent or legal guardian of the participants which were minors. In case the participant was younger than 16 years, the form had to be signed by a parent or legal guardian. In case the participant was 16 years or older, the parent or legal guardian was not required to return a signed version of the form. However, the form stated that parents or legal guardians could object to their child's participation via the contact details of the main researcher provided. The data handling and informed consent procedure used during the data collection campaigns was approved by the KU Leuven Social and Societal Ethics Committee under file number G-2021-3074.

Collected variables	Case 1: Secondary school	Case 2: Primary schools	Case 3: Test lecture room
<i>Environmental variables</i>			
Indoor temperature (°C)	X	X	X
Relative humidity (%)	X	X	X
Air velocity (m/s)			X
CO ₂ -concentration (ppm)	X	X	X
TVOC (ppb)			X
PM1 (µg/m ³)			X
PM2.5 (µg/m ³)			X
PM10 (µg/m ³)			X
Sound level (dB)	X	X	X
Lighting level (lx)	X	X	X
Specific enthalpy air (kJ/kg) *	X	X	X
Outdoor temperature (°C)	X		X
Running mean outdoor temperature (°C) *	X		X
* variable was calculated from monitored variables			
<i>Contextual variables</i>			
Clothing (Clo)	Simplified	Simplified	X
Location in room	X	X	X
Course type	X	X	X
Window opening status	X	X	
HVAC-system operation (Airflow rate, supply air temperature, etc.)			X

Table 1. Overview of collected variables during data collection campaigns.

Collected variables	Case 1: Secondary school	Case 2: Primary schools	Case 3: Test lecture room
Satisfaction IEQ (5-point rating)	X	X	X
Thermal sensation (7-point scale)			X
Thermal comfort perception (3-point scale)			X
IAQ perception (5-point scale)			X
Odour intensity (6-point scale)			X
Cause of dissatisfaction with IEQ	X	X	X

Table 2. Overview of the survey characteristics.

Overview collected variables. The measurement campaigns in the different case studies consisted of both continuous IEQ-monitoring and regular satisfaction assessments via RHRN surveys. The number of collected variables during the measurement campaigns differed between the cases. Additional thermal and IAQ instruments were used during the data collection campaign in the test lecture room (case 3). Due to the larger number of classrooms monitored in the secondary and primary school case studies, it was not practically feasible to use these devices in case studies 1 and 2. Consequently, the impact of variables such as air velocity, total volatile organic compounds (TVOC) and particulate matter (PM) concentration on occupant satisfaction can only be studied in case 3. An overview of the collected variables and survey characteristics are given in Tables 1 and 2, respectively. Note that in Table 2 a differentiation is made between “Thermal sensation” and “Thermal comfort”. The first referred to an evaluation of occupants’ thermal sensation on the ASHRAE 7-point thermal sensation scale (i.e., ranging from cold to hot) which is related to the PMV-PPD model²⁹. The latter resembled a simplified 3-point scale evaluating occupants’ thermal comfort perception (i.e., ‘uncomfortably cool’, ‘comfortable’ and ‘uncomfortably warm’). More details on the survey questions and scale can be found in the survey contents file at the data repository²⁸.

Selection and description of case studies. Data collection campaigns were performed in multiple classrooms of educational buildings in Flanders (Belgium). The classrooms were located in climate zone 4 A following the ASHRAE climate classification³⁰, representing a mixed-humid climate. The data collection campaigns were divided into three main case studies, i.e., three classrooms in a secondary school (Case 1), seven primary school classrooms (Case 2), and a test lecture room at KU Leuven Ghent Campus (Case 3). The three case studies were selected to cover a comprehensive range of classrooms. Consequently, the case studies varied in occupant (i.e., occupant age and education level), building (i.e., older and more recent buildings) and system (i.e., with and without ventilation system) characteristics. In addition, the selection was based on the practical feasibility to perform longitudinal measurement campaigns (e.g., motivation of occupants to participate and possibility to place environmental measurement devices).

The measurement duration in the cases varied from six weeks to six months. This period was sufficiently long to establish a useful dataset for further analysis. The measurement durations in the presented dataset is in line or longer than existing studies with durations of two weeks²³, three months¹² and six months¹⁷.

Case 1: Secondary school. Monitoring period. 21/04/2021 – 11/06/2021.

This period represents intermediate season conditions in Belgium.

Building and system characteristics. A secondary school in Haacht (Belgium) was selected as a first case study for this research. In total three classrooms were selected, which will be referred to as classrooms A, B and C (Fig. 2). Classroom C was located in an older school building dating back to 1961. This older building lacked a mechanical ventilation system, leading to classroom C being ventilated through manual opening of the windows. Hydronic radiators in the room provided heating for classroom C. Classrooms A and B were situated on the top floor of a new school building that finished construction in 2019. This recent school building consisted of 4 floors, basement included, and was equipped with a balanced mechanical ventilation system. This ventilation system had a total ventilation rate of 11500 m³/h and a rotary heat recovery with an efficiency of 81%. The ventilation flow rate per classroom was 690 m³/h and 25 m³/h per person. A constant air flow rate was provided by the ventilation system from 7 h until 18 h and was fully disabled outside these operation times. Classrooms A and B were each equipped with two inlet air openings. The extraction opening for classrooms A and B was situated in the hallway. Classrooms A and B had a similar layout and positioning of the inlet air openings. Furthermore, both classrooms were equipped with a floor heating system. The heating setpoint in classrooms A and B was set to 21.5 °C. None of the three classrooms was equipped with a cooling system. All classrooms were located at least 120 m away from a regional road. A detailed overview of the classroom properties is provided in Table 3.

Participants sample. In total 65 people participated in the survey campaign, of which 63 were students aged 14–17 years old. The two remaining participants were female teachers in the age category of 35–40 years old. The same group of occupants was questioned in the same classroom. Table 4 gives an overview of the demographics of the participating pupils.

Monitoring equipment. Three IEQ measurement devices (Renson Sense³¹) were placed in each classroom at different locations, i.e., in the front, middle and back of the classroom and at a height of approximately 1.5 m. The different measurement points made it possible to determine local IEQ conditions in the classroom and to detect whether any of the devices measured incorrectly. The devices measured the following IEQ-variables with a 5-minute timestep: temperature, relative humidity, light intensity, CO₂-concentration and sound level. The detailed properties of the measuring device can be found in Table 5. Furthermore, the outdoor air temperature was measured with a timestep of 10 minutes by a temperature sensor mounted on the North façade of the most recent school building at a height of 6 m.

Survey distribution. A survey planning was designed, based on the course schedules of the students, which proposed multiple moments to complete the survey. The same planning was used each week, since the course schedule remained unchanged during the monitoring period. We distinguished two types of completion moments, i.e., at the start and end of a course. The number of planned survey completion moments per classroom is shown in Table 6. The number of voting moments per week in classroom C is lower compared to classrooms A and B since the classroom C was less occupied by the selected group of students. The participants completed the online survey via their personal smartphone. A QR-code linking to the online survey was distributed in the classrooms to simplify the accessibility of the survey.

Case 2: Primary schools. The data collection campaign in case 2 was performed as part of a master thesis³².

Monitoring period. 21/10/2021 – 14/12/2021.

This period represents heating season conditions in Belgium.

Data collection was paused from 01/11/2021 until 07/11/2021 due to a school holiday.

Building and system characteristics. Seven classrooms from five different primary schools in Flanders (Belgium) were monitored for six weeks from October until December 2021. Each of these classrooms was occupied by the same group of pupils and teachers. Three of the classrooms (i.e. A, B and C) were identical and located in the same school. More detailed information on the classrooms is shown in Table 7.

Participants sample. A total of 127 pupils and 8 teachers participated at the data collection campaigns in the primary schools. Only one out of eight teachers were male. The age of the teachers varied from 23 to 52 years and was on average 32.1 years. Table 8 gives an overview of the gender and average age of the pupils per classroom.

Monitoring equipment. The classroom IEQ conditions were monitored using a HOBO-MX, HOBO U12-012 and Netatmo smart weather station. The three measurement devices were placed at opposite sides of each classroom and at different heights ranging from 0.8 to 1.5 m. The sampling frequency was set to 5 minutes. Table 9 gives an overview of the measured IEQ variables with the measurement accuracy per monitoring device.

During the monitoring period, all classrooms were equipped with an additional CO₂-monitoring device to inform the pupils and teachers about the IAQ condition in the classroom. The monitoring device was cased as

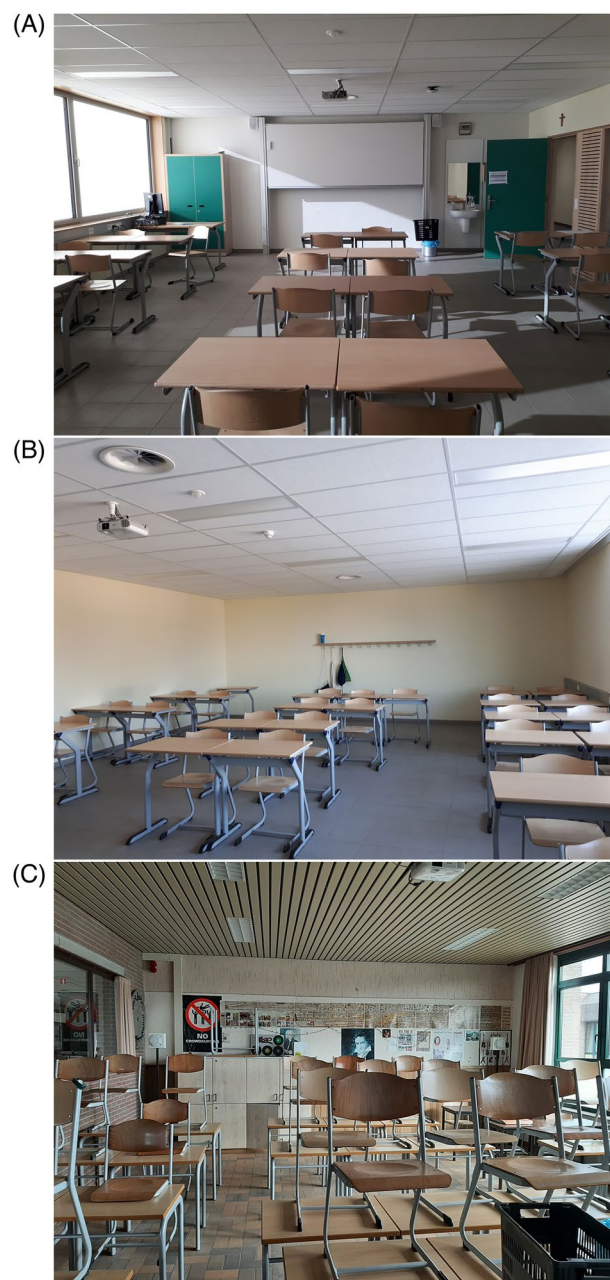


Fig. 2 Inside views of the secondary school classrooms A, B and C.

a bird in order to make it interesting for pupils (Fig. 3). The monitoring device had two functions, i.e., changing eye colours and whistling. The eye colour changed from green to blue and from blue to red at a CO₂-level of 800 ppm and 1200 ppm respectively. The device was set to whistle at 1000 ppm. The whistling function was only activated from the fourth week of the monitoring campaigns.

Survey distribution. A survey completion schedule was designed and sent to the participating schools. The participating classes were asked to complete the RHRN surveys five times a day for two days a week (i.e., Monday or Tuesday and Thursday or Friday). Of the five completion moments for the RHRN survey, three assessed the satisfaction upon entering the classroom (i.e., start of the day, after the morning break, after afternoon break), and two assessed the satisfaction when pupils and teachers were in the classroom for some time (i.e., before the morning break, end of the day). The two survey completion moments, i.e., upon entering and before leaving the classroom, were used to assess participants' satisfaction before and after adapting to the indoor environment. Two survey versions were distributed, i.e., a pupil and a teacher version. The pupil-version comprised of simpler language to make the survey more understandable for young pupils. The pupils completed the online survey via laptops of the schools that were available in the classrooms.

Properties	Classroom A	Classroom B	Classroom C
Floor area (m ²)	66	66	52.3
Volume (m ³)	175	175	164
Orientation	S	N	E
Window surface (m ²)	10.7	8.6	14.2
Airflow rate (m ³ /h)	690	690	/
Solar shading	External solar shading	/	Internal curtains
Window operability	Bottom-hung window	Bottom-hung window	Side-hung window
Heating system	Floor heating	Floor heating	Radiators
Sample size	26	22	17
Teachers	1	1	0

Table 3. Properties of the selected secondary school classrooms and sample size.

Classroom	Gender (male/ female/ unspecified)	Average age [year]
A	11 / 12 / 2	15.4
B	7 / 13 / 1	16.4
C	9 / 6 / 2	15.3

Table 4. Overview demographics of pupils in the secondary school classrooms.

Variable	Accuracy	Range
Air temperature	±0.2 °C (range 0–60 °C)	–10–60 °C
Relative humidity	±2% (range 10–90%)	0–100%
Light intensity	/	0–1000 Lux
CO ₂ concentration	±5% + 40 ppm (range 400–2000 ppm)	0–5000 ppm
Sound level	±1 dB	29 dB–120 dB

Table 5. Properties of the IEQ measurement device deployed in the secondary school classrooms.

Classroom	Total moments / week	Beginning of course	End of course
A	11	4	7
B	12	4	8
C	6	3	3

Table 6. Survey completion moments per the secondary school classroom.

Classroom	Volume	Operable window area + orientation	Ventilation system
A	182 m ³	1 m ² East	Mechanical extraction
B	182 m ³	1 m ² East	Mechanical extraction
C	182 m ³	1 m ² East	Mechanical extraction
D	216 m ³	11.77 m ² North-West	No mechanical ventilation
E	380 m ³	4 m ² South-East	No mechanical ventilation
F	171 m ³	1 m ² West	Balanced mechanical
G	165 m ³	1.76 m ² South-East	No mechanical ventilation

Table 7. Primary school classroom properties.

Furthermore, pupils' motivation for filling in the survey and their interest in the study were assessed at the end of a school day. From the fourth week of the monitoring campaign, the RHRN survey was extended with two questions. One question asked about the whistling of the CO₂-monitoring device. In a second question, the pupils could indicate which actions he/she or the teacher took to improve the IAQ in the classroom.

Case 3: Test lecture room (TLR) – KU Leuven Ghent campus. Monitoring period. 30/11/2021 – 25/05/2022.

This period represented heating and intermediate season conditions in Belgium.

Measurements ran over three different experiment campaigns. The first ran from 30/11/2021 to 20/01/2022. The second campaign was performed during a full academic semester from 14/02/2022 to 19/05/2022. The last

Classroom	Gender (male/ female/ unspecified)	Average age [year]
A	10 / 12 / 1	9.6
B	9 / 9 / 1	9.9
C	8 / 12 / 2	10.0
D	4 / 12 / 2	10.8
E	3 / 6 / 6	11.0
F	10 / 9 / 0	10.8
G	6 / 2 / 3	10.4

Table 8. Overview demographics of pupils in primary school classrooms.

Device	Variable	Accuracy
Netatmo smart weather station (indoor module)	Air temperature [°C]	±0.3 °C
	Relative humidity [%]	±3%
	CO ₂ -concentration [ppm]	±50 ppm (<1000 ppm) 5%(>1000 ppm)
	Air pressure [hPa]	±1 mbar
	Sound level [dB]	/
Onset HOBO MX	Air temperature [°C]	±0.21 °C
	Relative humidity [%]	±2% (20%–80%) ±6% (<20% & >80%)
	CO ₂ -concentration [ppm]	±50 ppm
Onset HOBO U12-012	Air temperature [°C]	±0.35 °C
	Relative humidity [%]	±2.5% (10%–90%) ±5% (<10% & >90%)
	Light intensity [Lx]	/

Table 9. Properties of the IEQ measurement devices in primary schools.



Fig. 3 Example of the CO₂ monitoring device⁴⁴ that present during the data collection campaigns in the primary school classrooms.

experiment campaign consisted of six consecutive study sessions on Tuesday afternoons from 13:30–17:30 from 19/04/2022 to 25/05/2022.

Building and system characteristics. Case study three was situated in a nearly-zero energy (nZEB) educational building, located in Ghent, Belgium. The educational building³³ comprised of two lecture rooms used by university and college students, a staircase and a technical room. The two lecture rooms were situated above each other and were identical in dimensions and in the positioning of ventilation openings. The lecture rooms differed in envelope composition. The lecture room on the 1st floor had a brick external wall with exterior insulation, an air tightness of 0.41 h⁻¹ at 50 Pa, while the 2nd floor lecture room had a lightweight timber frame external wall with the same U-value (0.15 W/m².K) and an airtightness of 0.29 h⁻¹ at 50 Pa. The roof and floor have U-values of 0.14 and 0.15 W/m².K respectively. The building had triple-glazed windows, with a U-value of 0.65 W/m².K, on the southwest and northeast façades, having a g-value of 0.52, a window-to-wall ratio (WWR) of 27% and a window-to-floor ratio (WFR) of 13%. The windows were equipped with internal and external solar shading

consisting of screens. The external screens were located on the southwest windows and closed automatically when the incident solar radiation exceeds 250 W/m^2 , however, this control could be manually overruled by the users.

The building was equipped with a variable air volume (VAV)-based all-air HVAC system. The ventilation flow rate and supply air temperature were demand-controlled based on a room temperature and CO_2 -setpoint. The all-air system had a maximum air flow rate of $2200 \text{ m}^3/\text{h}$ per lecture room leading to $27.5 \text{ m}^3/\text{h}$ per person at a maximum occupancy of 80 occupants. The air was supplied to the space via a displacement ventilation system (four diffusers in the corners) and exhausted via a grid at the back of the lecture room situated at a height of 2 m. The lecture rooms had a floor area of 140 m^2 and a volume of 380 m^3 .

The building was cooled by three techniques of ventilative cooling: (1) natural night ventilation with the windows serving as supply and exhaust openings, (2) modular bypass in the air handling unit (AHU), (3) indirect evaporative cooling (IEC) with a maximum capacity of 13.1 kW . As for heating, the air was pre-heated by air-to-air heat recovery via two crossflow plate heat exchangers connected in series with an efficiency of 78%. Furthermore, heating coils with a capacity of 7.9 kW were installed in the supply ducts of each lecture room. The two lecture rooms served as a living lab for the *Building Physics and Sustainable Design* research group at KU Leuven. The test lecture rooms were thermally insulated from each other. The lower lecture room was used as the case study classroom. We referred to the classroom as the test lecture room (TLR). An inside view of the TLR is shown in Fig. 4. More information on the TLR can be found in the building descriptive paper³³.

During the experimental campaigns, the room temperature and CO_2 setpoints were varied in order to expose the participants to different conditions. The room temperature setpoint was changed between 18°C to 26°C . Originally, the classroom used the setpoint of 23°C . Furthermore, two different CO_2 -setpoints were set, i.e., 500 ppm (assuming maximum ventilation flow rate during occupancy) and 1000 ppm.

Participants sample. The group of participants varied between the three experiment campaigns. During the first experimental campaign, the participants comprised of a group of international students who had courses in the TLR classroom for a full semester. Thirty-nine of these students participated in the data collection campaign. Students were mostly a whole day in the classroom for five days a week. The second experimental campaign consisted of two groups of Belgian bachelor students of which 65 participated in the data collection campaign. The third experimental campaign included 17 Belgian volunteers who were recruited via a flyer and poster campaign at the university campus. The 17 volunteers participated at six study session where they could work or study while being present in the TLR. An overview of the participants' gender and age is given in Table 10.

Monitoring equipment. The TLR classroom was equipped with multiple sensors and a weather station to measure both indoor and outdoor environmental conditions with a timestep of 1 minute. Moreover, an IAQ sensor (Ethera NEMO³⁴) was placed in the middle of the classroom that had a sampling frequency of 10 minutes. Furthermore, a thermal comfort meter (Testo 480³⁵) was placed in the classroom to monitor additional properties of the thermal environment in the classroom (i.e., air velocity and mean radiant temperature) with a timestep of 1 minute. The measured variables and their accuracies are shown in Table 11.

Survey distribution. An English version of the survey was provided for the international participants. The survey completion schedule differed between the experimental campaigns. During the first experimental campaign, a survey completion schedule was proposed to the participants based on their courses in the TLR. Participants of the second experimental campaign were asked to complete the survey upon entering and leaving the TLR. Consequently, the survey completion frequency depended on how often the students had courses in the TLR, i.e., one to four times a week. The third experimental campaign aimed to maximise the number of survey responses per occupants. Therefore, the participants of the third experimental campaign were asked to complete the RHRN survey upon entering and leaving the TLR when attending a study session. In addition, a 30-minute timer was active during the study sessions which informed participants to complete the RHRN survey when the timer finished. Participants completed the online survey using their personal smartphone or laptop. To facilitate access to the online survey, QR codes linking to the survey were distributed to participants.

The RHRN surveys also comprised of questions assessing the study engagement of the participants. In addition, participants of the second and third experimental campaign completed two cognitive tests to evaluate their memory and processing speed. Results of the study engagement and cognitive performance of the participants can be found in a previous publication³⁶.

Dataset construction. The collected survey data, including the satisfaction assessments, was combined with IEQ, outdoor climate and HVAC system-related variables into one dataset. The responses from the RHRN survey were merged with the nearest environmental measurement in time of the IEQ sensors or weather station. The data was merged based on the nearest measurement in time since the RHRN survey evaluated occupants' satisfaction with the IEQ conditions at the time of completing the survey. After the data was merged, a data cleaning process was performed which comprised of deleting incomplete survey responses from the dataset. A survey response was incomplete when an assessment of the occupant's satisfaction was lacking. No further data cleaning was performed, but the authors recommend checking the time between the survey response and a measured environmental variable. In cases where the time difference exceeds a chosen threshold (e.g., 15 or 30 minutes), the value of the environmental variable should be set to NaN. As a result, only corresponding environmental conditions are considered in future analyses on the datasets.



Fig. 4 Inside view of the test lecture room.

Experimental campaign	Gender (male/ female/ unspecified)	Average age [year]
1	7 / 23 / 9	26.0
2	50 / 10 / 5	20.3
3	12 / 4 / 1	22.9

Table 10. Overview of participants' demographics for the TLR case.

Source	Variable	Accuracy
Weather station	Outdoor temperature	0.3 °C at 20 °C
	Outdoor relative humidity	0.1%
Building management system	Indoor room temperature	0.1 °C
	Relative humidity room	1%
	CO ₂ -concentration	30 ppm +3% of reading
	Airflow rate	0.04%
IAQ sensor	TVOC (Aldehyde, BTEX, NH ₃ , H ₂ S, hydrocarbon)	3 ppb
	PM ₁ / PM _{2.5} / PM ₁₀	10 µg/m ³
Comfort meter	Air temperature	0.5 °C
	Mean radiant temperature	0.1 °C
	Relative humidity	2%
	Air velocity	0.01 m/s

Table 11. Properties monitoring equipment TLR.

Data Records

Summary. All data records have been uploaded to KU Leuven research data repository (RDR) and include four datasets files, a parameter description file, a survey description file and metadata²⁸. A total of 6834 satisfaction assessments from 321 occupants were collected throughout the data collection campaigns in the three case study classrooms after data cleaning. Figure 5 gives an overview of the number of collected satisfaction assessments and participants per case study

Contents. This section describes the different files in our datasets.

FullDataset_SecondarySchool.csv. This datafile consists of 834 rows in which each row represents an occupant survey response with the measured environmental conditions in the classroom for case 1. The file format is a UTF-8 comma-separated csv-file. This datafile comprises of 86 columns each representing a different parameter (e.g., timestamp, survey response, environmental measurement, ...). A detailed overview and definition of each parameter can be found in the "Dataset_ParameterDescription.xlsx"-file.

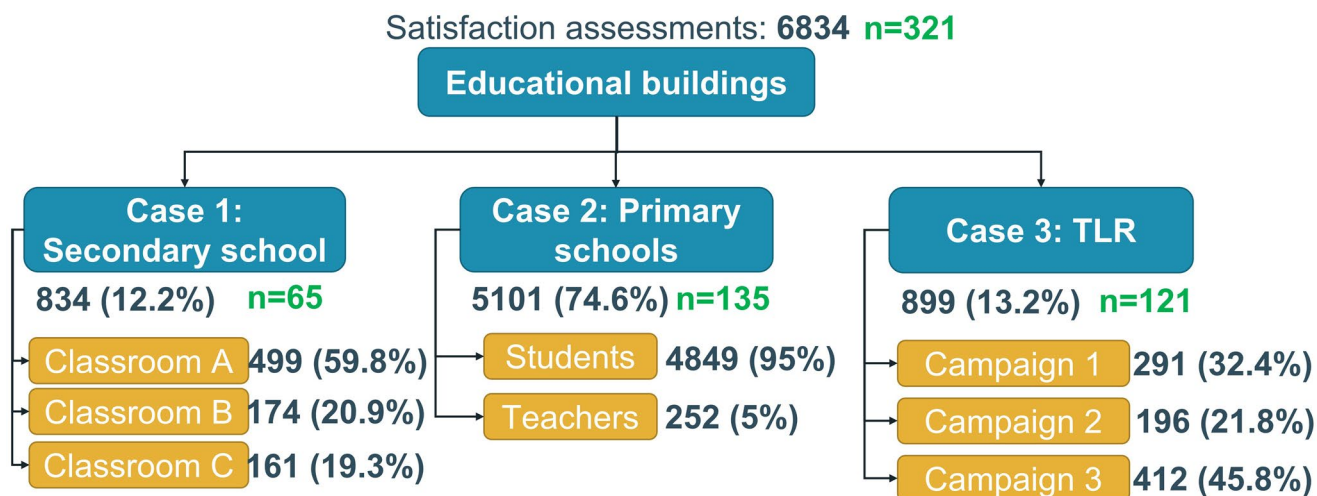


Fig. 5 Overview of the satisfaction assessments that are collected per case study. (Number of unique participants **n** are shown in green).

FullDataset_PrimarySchools.csv. This datafile consists of 5101 rows in which each row represents an occupant survey response with the measured environmental conditions in the classroom for case 2. The file format is a UTF-8 comma-separated csv-file. This datafile comprises of 86 columns each representing a different parameter (e.g., timestamp, survey response, environmental measurement, ...). A detailed overview and definition of each parameter can be found in the “Dataset_ParameterDescription.xlsx”-file.

FullDataset_TestLectureRoom.csv. This datafile consists of 899 rows in which each row represents an occupant survey response with the measured environmental conditions in the classroom for case 3. The file format is a UTF-8 comma-separated csv-file. This datafile comprises of 145 columns each representing a different parameter (e.g., timestamp, survey response, environmental measurement, ...). A detailed overview and definition of each parameter can be found in the “Dataset_ParameterDescription.xlsx”-file.

FullDataset_CombinedSimple.csv. This datafile consists of 6834 rows in which each row represents an occupant survey response with the measured environmental conditions in the classroom for all cases. The file format is a UTF-8 comma-separated csv-file. This datafile only includes a set of key variables that were measured in all three cases (e.g., occupant satisfaction scores, room temperature, relative humidity,...). Therefore, the datafile comprises of only 59 columns each representing a different parameter (e.g., timestamp, survey response, environmental measurement, ...). A detailed overview and definition of each parameter can be found in the “Dataset_ParameterDescription.xlsx”-file.

Dataset_ParameterDescription.xlsx. This file provides a description of all the parameters (i.e., column headers) in the four datafiles: “FullDataset_SecondarySchool.csv”, “FullDataset_PrimarySchools.csv”, “FullDataset_TestLectureRoom.csv” and “FullDataset_CombinedSimple.csv”. The datafile has an xlsx-format and comprises of four tabs, i.e., one for each datafile.

OccupantSurveys.pdf. This file provides a description of the occupant survey structure and content used in each of the three cases. The document provides an overview of all survey questions and response scales in the original survey language of Dutch and translated into English. The file has a pdf-format.

README.txt. This file “README.txt” provides the necessary metadata for the dataset. The file has a txt-format.

Technical Validation

Survey validity. The survey questions and scales were tested via pilot tests for comprehensibility to ensure that the surveys assessed what they were intended to assess. The pilot tests were performed at the start of the data collection campaigns. During these pilot tests, the surveys ended with an optional open text field where participants could provide feedback on the survey content. Participant feedback was checked and used to adapt the survey and increase its comprehensibility. The pilot tests did not indicate any misunderstanding of the survey contents. In addition, the surveys that were distributed to minors (i.e., in secondary and primary schools) were reviewed by teachers to assess the comprehensibility of the surveys for the target age group of participants.

Measurement accuracy. The devices and sensors used to measure the environmental conditions are described in the Methods section. Multiple measurement devices were placed in the monitored classrooms to determine local IEQ conditions in the classroom and to detect whether any of the devices measured incorrectly.

In addition, the measurement devices were positioned in locations that replicated the indoor environmental conditions experienced by the participants.

Since different measurement devices were used over the different case studies, differences in sampling frequencies of the measurements occurred (i.e., one, five and ten minutes). During the dataset construction, the occupants' survey response is merged with the environmental measurement that is nearest in time (see **Dataset construction** section). Consequently, the difference in sampling frequencies is mitigated in the final dataset.

Potential data losses were indicated in the dataset by NA. In case 1, no environmental measurements were missing. In case 2, 999 lighting level measurements were missing due to a malfunction of the lighting measurement device in one of the monitored classrooms. In case 3, 29 datapoints lack a measurement of the indoor temperature, relative humidity and CO₂-concentrations due to a short malfunction of the building monitoring system. In addition, 37 datapoints lack a measurement of TVOC- and PM concentrations. Moreover, air velocity was not measured for 63 datapoints. Finally, the lighting and sound level measurements (487 and 581, respectively) were missing since these variables were not measured for the full data collection campaign.

Occupant satisfaction assessments. Figure 6 shows the dissatisfaction rates per case study and per evaluated IEQ domain. The dissatisfaction rate was the percentage of satisfaction assessments with a score lower than three on the five-point rating scale. The results show that occupants were often dissatisfied with the thermal and IAQ conditions in their classrooms. However, differences between the case studies were present. In the secondary school, thermal and IAQ dissatisfaction rates were around 20% making it the highest of all evaluated domains. In contrast, occupants in the primary schools were mostly dissatisfied with the acoustical conditions (16%) in their classroom followed by the IAQ conditions (14%). In the test lecture room (case 3), the occupants showed the highest dissatisfaction with the thermal (13%) and visual environment (9%). A more detailed distribution of the thermal and IAQ satisfaction scores are shown per case in Figs. 7 and 8. These figures show that the peak in thermal and IAQ satisfaction scores was at 3 for the secondary school case while it was at 4 for the TLR case. The primary school case had a similar frequency of satisfaction score of 3, 4 and 5, thus not showing a clear peak as in the other two case studies.

The satisfaction assessments showcase an imbalance in the dataset, i.e., a larger proportion of assessments indicate satisfaction than dissatisfaction. However, imbalanced datasets are common in field data collection campaigns and are representative for real-life situations. Previous studies evaluating predictive models using field measurement data also faced imbalance, e.g., evaluated predictive thermal comfort models on more imbalanced data, e.g., “No change” (58%), “Cooler” (35%), “Warmer” (7%)¹⁷ and “No change” (70%), “Want cooler” (17%), “Want warmer” (13%)³⁷. In buildings without specific issues (e.g. HVAC system malfunctions), it is expected that a majority of votes will indicate comfortable conditions.

Measured environmental conditions. A selection of three key environmental variables is investigated in more detail, i.e., room temperature, relative humidity and CO₂-concentration. The descriptive statistic of each variable is visualised per case study. In addition, outlier detection was performed on for each set of environmental variables. A datapoint was considered an outlier when its value fell outside the range between the third quartile plus 1.5 times the interquartile range and the first quartile minus 1.5 times the interquartile range.

Figure 9 gives an overview of the values for room temperature in the datasets for all three cases. For case 1 (i.e., secondary school), the descriptive statistics shows that the measurement data comprised mostly of indoor temperature which tended to be high (i.e., average temperature of 23.9°C). 37 measured values for room temperature were classified as an outlier as they exceeded a temperature of 26.8°C. The maximum value measured for room temperature in case 1 was 28.7°C. These results suggest that a short period of overheating occurred during the measurement campaign in case 1. On the contrary, the room temperature in measured in the primary school classrooms (case 2) was lower with an average temperature of 18.9°C. 32 outliers were detected for room temperature as they were lower than 14°C. These low room temperatures occurred due to a short malfunction of the heating system in one of the monitored classrooms. Case 3 comprised of room temperature measurements ranging from cooler to warmer, i.e., 17.9°C to 27.2°C. The average room temperature measured in case 3 was 22.8°C, and no outliers were detected.

Figure 10 gives an overview of the values for relative humidity in the datasets for all three cases. Low relative humidities were measured in case 1 with a minimum value of 19.6% and an average value of 35.4%. No values for relative humidity were categorised as outlier in case 1. On the contrary, eight data points were determined outliers in case 2 as they exceeded the value of 63.5%. The values measured for relative humidity were higher in case 2 with an average value of 45.9%. In case 3, an average relative humidity value of 39.4% was measured. Additionally, five values were identified as outliers due to exceeding a threshold of 58.1%.

Figure 11 gives an overview of the values for CO₂-concentration in the datasets for all three cases. In case 1, an average CO₂-concentration of 720 ppm was measured which reached a maximum value of 1243 ppm. No values for CO₂-concentration were detected as outlier in case 1. On average, the measured CO₂-concentrations were lower with a mean value of 616 ppm. A total of 202 CO₂ concentration values exceeded 980 ppm in case 2 and were therefore categorised as outliers. The average of the measured CO₂-concentrations in case 3 was 652 ppm. 100 values were categorised as outlier as they exceeded a CO₂-concentration of 881 ppm. Many of these outliers were measured during a brief period when that the demand-controlled ventilation (DCV) system was not operational during the data collection campaign. During this malfunction, the CO₂-concentration reached a maximum value of 2201 ppm.

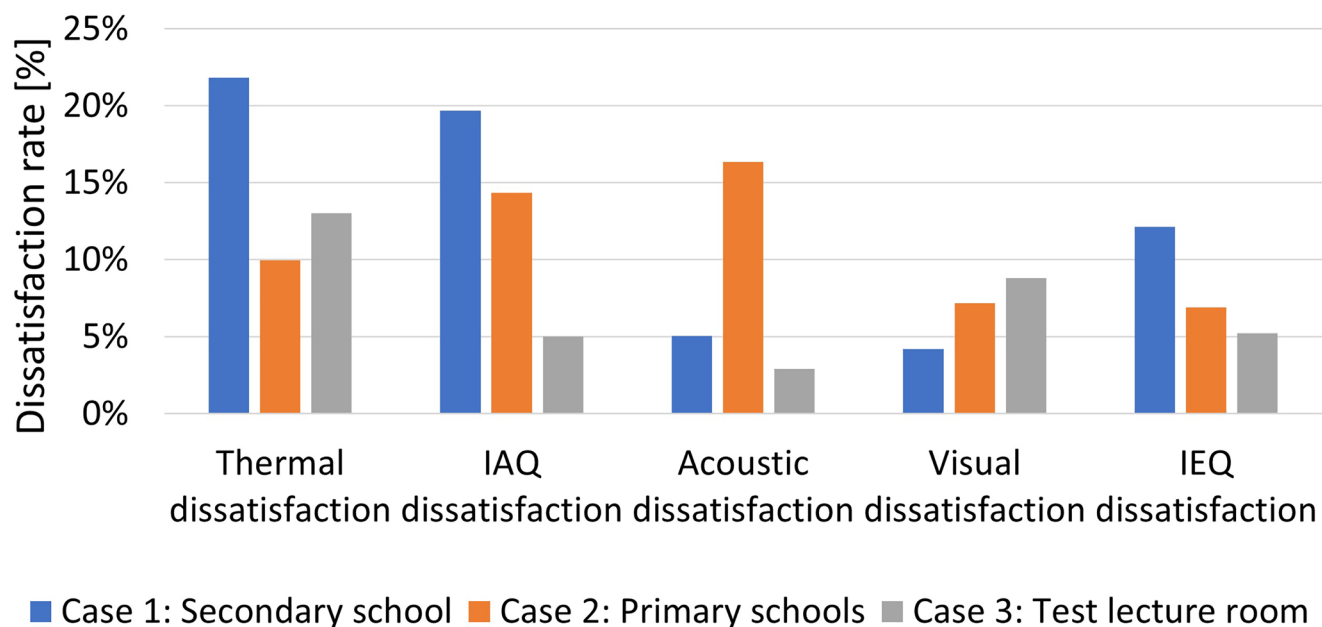


Fig. 6 Dissatisfaction rates per IEQ domain for the three case studies. The dissatisfaction rate represents the percentage of satisfaction assessments with a score lower than three on the five-point rating scale.

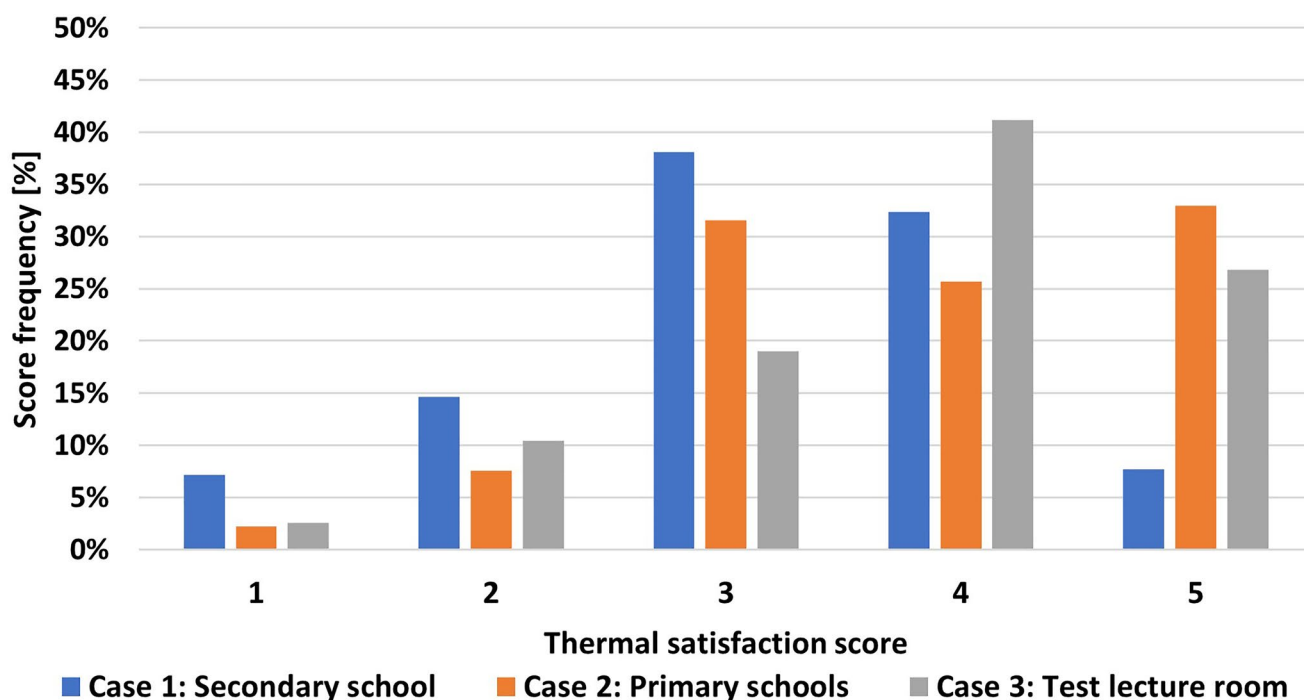


Fig. 7 Distribution of the thermal satisfaction scores (1–5) in the three case studies shown as percentage of the total number of satisfaction assessments.

Usage Notes

COVID-19 measures. The data collection campaigns in the secondary (case 1) and primary schools (case 2) were performed when COVID-19 measures were still active in Belgian schools. Consequently, occupants were required to wear face masks during the data collection campaigns. Wearing a face mask could have impacted occupants' perception of the IEQ. Few studies have investigated the impact of face masks on human perception of the IEQ. One such study, by Lang *et al.*³⁸, showed that occupants' thermal comfort, thermal acceptability and perceived air quality decreased over time when present in an office space with a room temperature and relative

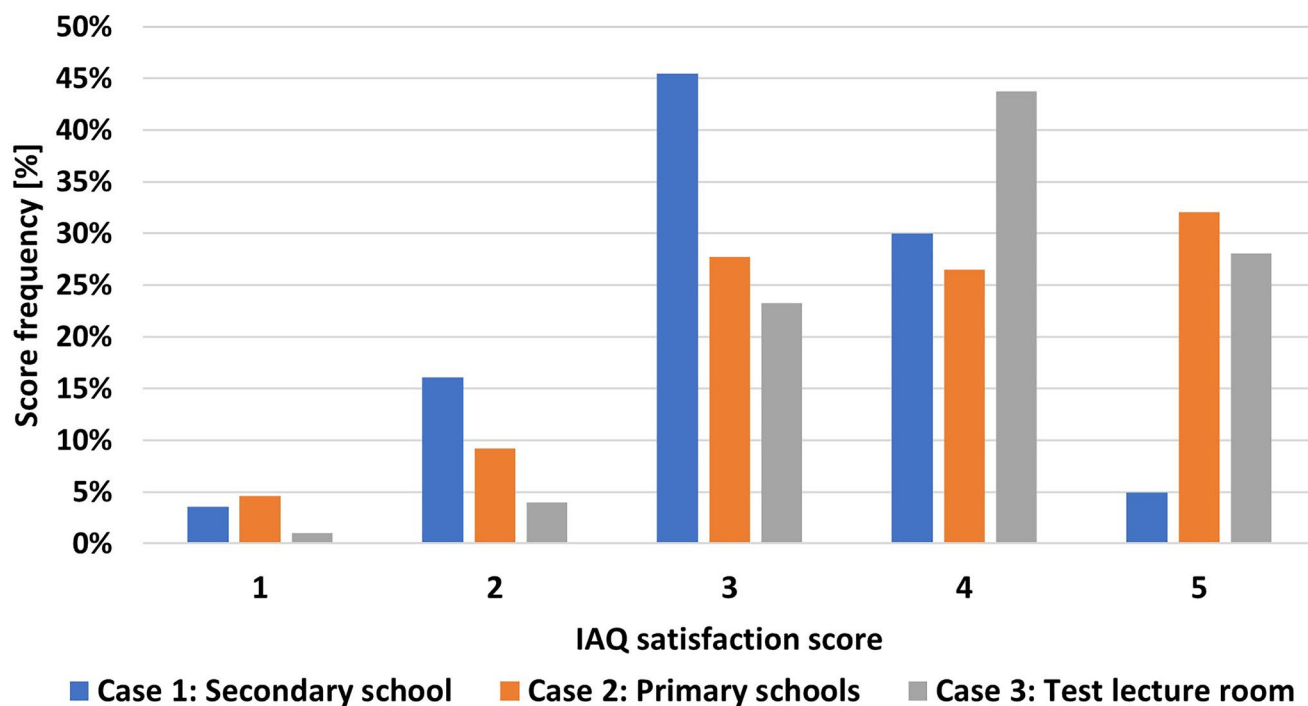


Fig. 8 Distribution of the IAQ satisfaction scores (1–5) in the three case studies shown as percentage of the total number of satisfaction assessments.

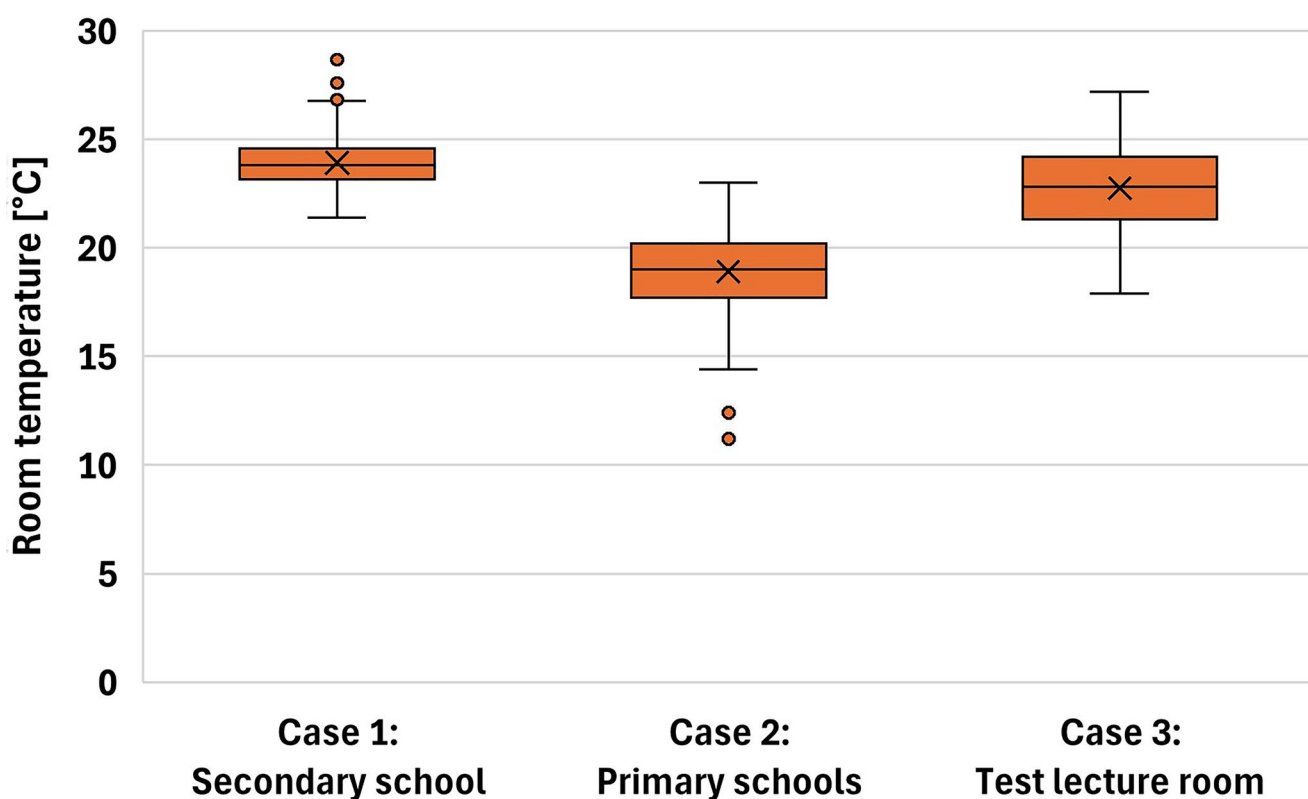


Fig. 9 Boxplot of the measured room temperatures in the datasets for the three case studies: Secondary school (Case 1), Primary schools (Case 2) and Test lecture room (Case 3). The dots represented values that were categorised as outliers.

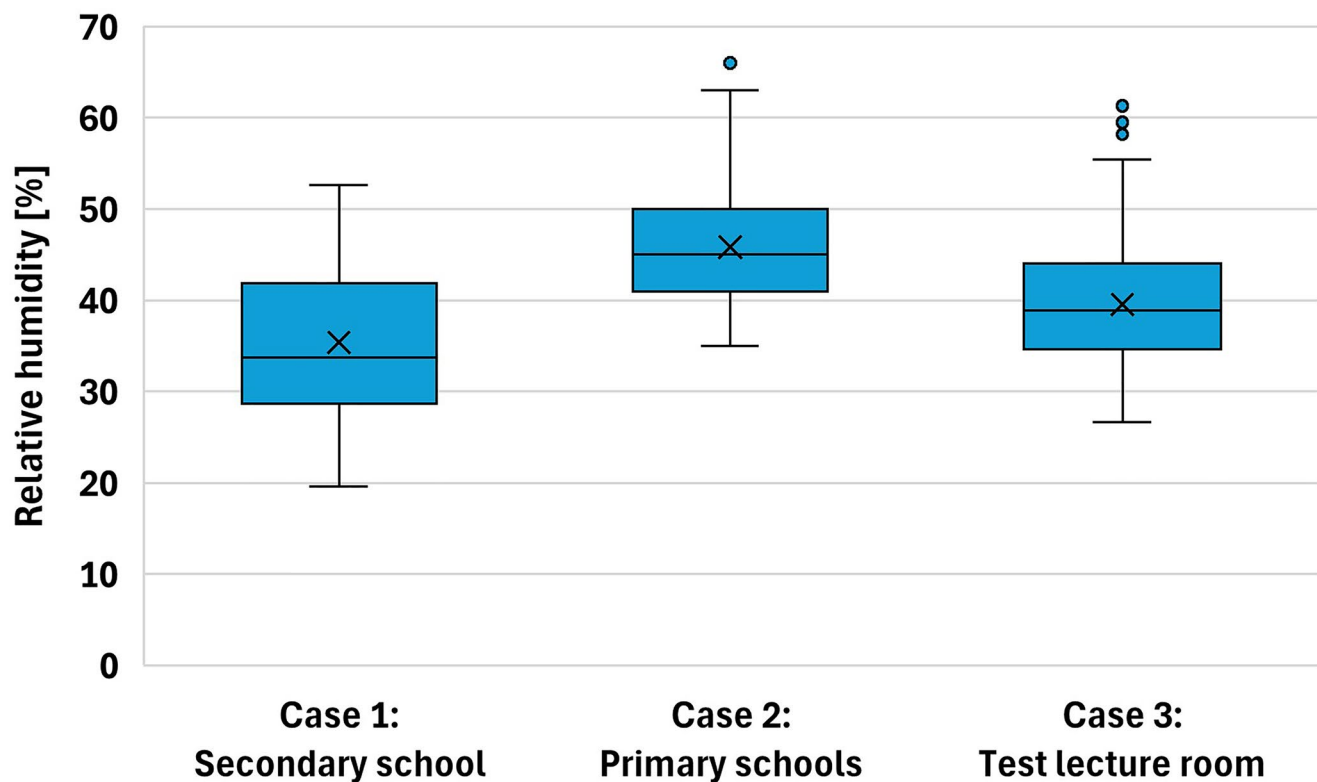


Fig. 10 Boxplot of the measured relative humidity in the datasets for the three case studies: Secondary school (Case 1), Primary schools (Case 2) and Test lecture room (Case 3). The dots represented values that were categorised as outliers.

humidity of around 23 °C and 40%, respectively, while wearing face masks. Another study by Tang *et al.*³⁹ found minor difference in thermal comfort between students with and without face masks in a university library. The authors quantified that the preferred operative temperature of occupants with a face mask was 0.5 °C lower than occupants without a face mask³⁹. On the contrary, a third study did not find a significant impact of wearing a face mask on occupants' thermal comfort that were present in climate chamber conditioned at 24 °C, 26 °C and 30 °C⁴⁰. To conclude, robust and conclusive evidence on the impact of face masks on occupants' IEQ perception is still missing.

In addition to the obligation to wear face masks, schools in cases 1 and 2 were advised to open windows as much as possible as part of the measures to combat the spread of the COVID-19 virus. As a result, the window opening behaviour differed from what was expected during heating season with at least one or more windows being opened during occupancy. The impact of the window opening will be reflected in the measured indoor conditions, such as a lower indoor temperature and CO₂-concentration. Consequently, the conditions during these measurement campaigns deviated from what would be expected under normal conditions (i.e., in the absence of active regulations to reduce the risk of respiratory infection).

The active COVID-19 measures may lead to atypical conditions. However, given the proven effectiveness of face masks in reducing the transmission of respiratory diseases, their use can be expected to become a standard preventive measure in future scenarios such as seasonal flu outbreaks or emerging epidemics. Future researchers using the dataset should be aware of the contextual conditions of the data collected in the secondary and primary school classrooms due to the active COVID-19 measures. Specifically, findings based on these case studies may face limitations in generalising to non-pandemic conditions. Nevertheless, COVID-19 measures can be treated as a confounding contextual factor when analysing this dataset alongside non-pandemic data.

Outdoor climate conditions. The current dataset is limited to measurements performed in educational buildings located in the region of Flanders (Belgium). As a result, all measurements were collected under the same climate condition, ASHRAE climate zone 4 A (mixed-humid climate)³⁰. The climatic homogeneity of the current datasets limits the generalisability of findings to other regions with different climate zones. Therefore, if users aim to develop and validate a multi-climate comfort model, the current dataset should be supplemented with additional data from different climate zones.

Data availability

The dataset is available at KU Leuven Research Data Repository: <https://doi.org/10.48804/CZQNUU>.

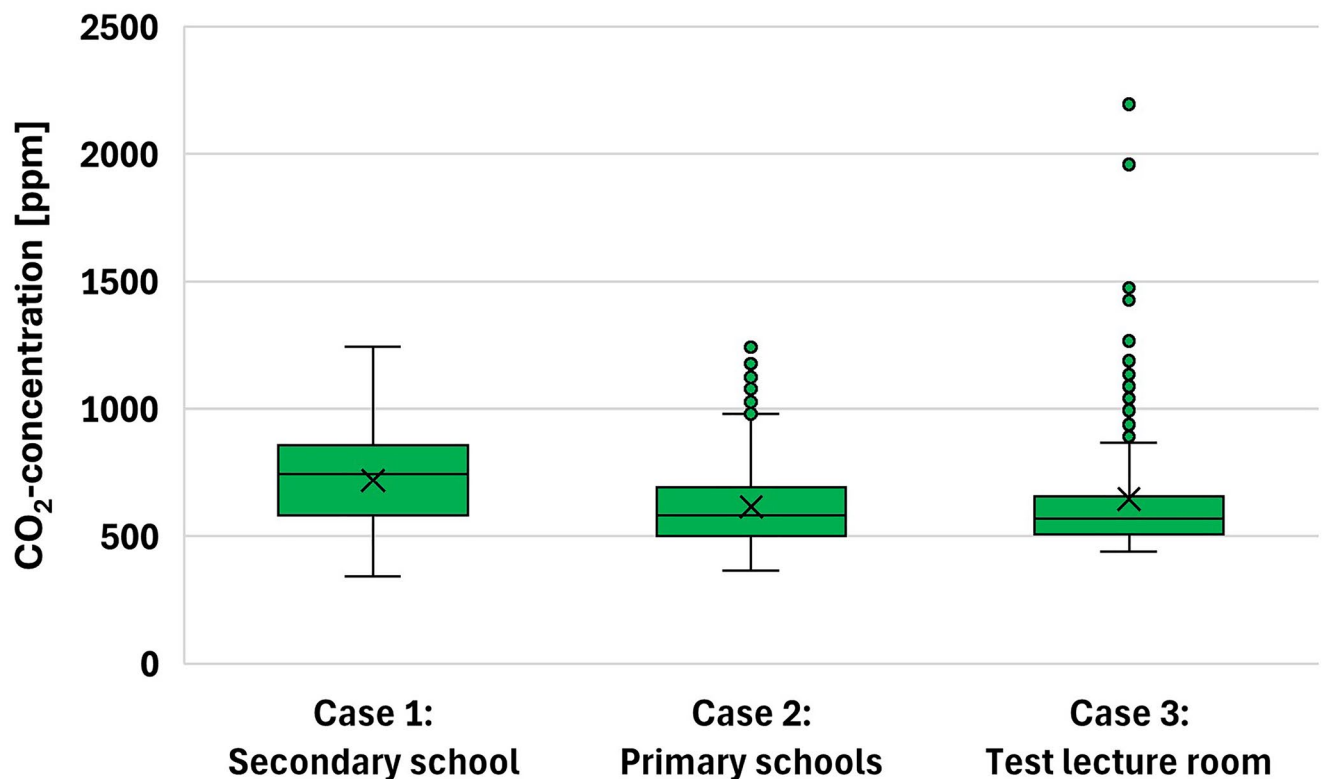


Fig. 11 Boxplot of the measured CO₂-concentrations in the datasets for the three case studies: Secondary school (Case 1), Primary schools (Case 2) and Test lecture room (Case 3). The dots represented values that were categorised as outliers.

Code availability

Additional variables were calculated and added to the published datasets. The air enthalpy (EA) and running mean outdoor temperature (T_{rm}) were added using the relevant functions in the “pythermalcomfort” package⁴¹ for Python. EA was added to the datasets because previous studies have shown its relationship with occupants’ IAQ perception^{42,43}. Trm was included in the datasets since it is part of the adaptive thermal comfort model¹¹. The “pythermalcomfort” package⁴¹ is an open-source toolkit to compute thermal comfort indices in Python.

The study did not use any custom code to generate data.

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Author contributions

Quinten Carton designed the methodology, performed the data collection campaigns, constructed the datasets, prepared and edited the paper draft. Jakub Kolarik designed the methodology, supervised the project, reviewed and edited the paper draft. Hilde Breesch designed the methodology, supervised the project, reviewed and edited the paper draft, and acquired funding.

Competing interests

The authors declare no competing interests.

Additional information

Correspondence and requests for materials should be addressed to Q.C.

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